Yiming Zhang

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EDUCATION

Fudan University | Shanghai, China
M.S. in Information and Communication Engineering. Advisor: Prof. Feng Wang — Sep 2025 – Present

Fudan University | Shanghai, China
B.E. in Intelligence Science and Technology. Advisor: Prof. Feng Wang — Sep 2021 – Jul 2025

RESEARCH INTERESTS

My research focuses on adversarial robustness and security in machine learning, with an interdisciplinary emphasis on synthetic aperture radar (SAR) imagery, whose physics-driven formation process poses distinctive challenges. I have published research on adversarial attacks against SAR classification and detection models and am currently developing tailored adversarial training strategies for SAR semantic segmentation. Meanwhile, I am particularly interested in the security, robustness, and interpretability of large language models (LLMs) and vision-language models (VLMs), while remaining open to broader research directions in machine learning and computer vision.

PUBLICATIONS

Y. Zhang, W. Qin, Y. Liu and F. Wang, "SRAW-Attack: Space-Reweighted Adversarial Warping Attack for SAR Target Recognition," in IEEE Geoscience and Remote Sensing Letters, 2026. [Code]

Y. Zhang, W. Qin and F. Wang, "Towards Physically Realizable Adversarial Attenuation Patch against SAR Object Detection," 2026 China Aerospace Information Technology Conference (CAIT), Tongxiang, China, 2026. [Code]

Y. Liu, W. Qin, **Y. Zhang** and F. Wang, "Cross-Domain Universal Adversarial Perturbations for Synthetic-to-Measured SAR Target Classification Attacks," in IEEE Geoscience and Remote Sensing Letters, 2026.

W. Qin, **Y. Zhang** and F. Wang, "Rethinking the Optimal Strategy of Deep Learning for Sar Image Semantic Segmentation," 2023 SAR in Big Data Era (BIGSARDATA), Beijing, China, 2023.

AWARDS AND HONORS

- **National Scholarship (top 1%)** *2026*
- **Fudan University Academic Scholarship for Graduate Students** *2026*
- **Fudan University Wang-dao Scholar (Undergraduate Research Program)** *2025*
- **Fudan University Scholarship for Outstanding Undergraduate Students** *2025*
- **Mu Xin AI Scholarship at SIST, Fudan University** *2024*
- **Fudan University Scholarship for Outstanding Undergraduate Students** *2023*

SELECTED PROJECTS AND COMPETITIONS

National Natural Science Foundation of China (NSFC) General Program — Sep 2026 – Present
Key Contributor | PI: Prof. Feng Wang

- Contributed to research ideation and grant proposal writing.
- Developed a novel adversarial attack method and conducted research on a tailored adversarial training strategy for DNN-based SAR image interpretation, advancing two core research tasks within the program.

Wang-dao Program: Adversarial Example Design for SAR Object Detection — Aug 2023 – Mar 2025
Independent Researcher | Advisor: Prof. Feng Wang

- Explored the learned representations and behavior of DNNs trained on SAR imagery, revealing how shifts in the training domain affect the adversarial vulnerability.

- Designed a novel patch-based adversarial attack method for SAR object detection, incorporating the characteristics of SAR imagery.
- Evaluated attack effectiveness and cross-model transferability against baseline adversarial patch methods.

Raytron Competition: Large Model-based Infrared Image and Scene Generation Aug 2025 – Nov 2025
Team Leader and Core Participant

- Utilized the Stable Diffusion model and ControlNet to generate a complete scene containing a specified target slice, with key innovations in fine-tuning dataset construction and visual prompt design.
- Proposed a post-processing pipeline to integrate and animate static target patches in Open-Sora-generated infrared background videos, improving motion consistency and visual quality.
- Awarded the national first prize, receiving a cash prize of RMB 40,000.

ACADEMIC AND EDUCATIONAL ENGAGEMENT

2026 China Aerospace Information Technology Conference (CAIT) May 2026
Wuzhen, Tongxiang, China
Poster Presentation: Towards Physically Realizable Adversarial Attenuation Patch against SAR Object Detection

Teaching Assistant: Introduction to Aerospace Information and Technology Spring 2026
Fudan University, Shanghai, China

SKILLS

Languages: English (Professional Working Proficiency), Mandarin (Native)
Programming Languages: Python / C++ / C / MATLAB / PyTorch
Development and Research Tools: LaTeX, VS Code, Git, Markdown, Codex, etc.